

Interaction Prototyping With Video: Bridging Video Interaction Analysis & Design

Hannah Pelikan
hannah.pelikan@liu.se
Linköping University
Linköping, Sweden

Leelo Keevallik
leelo.keevallik@liu.se
Linköping University
Linköping, Sweden

Yoyo Tsung-Yu Hou
th588@cornell.edu
Cornell University
Ithaca, NY, USA

Mathias Broth
mathias.broth@liu.se
Linköping University
Linköping, Sweden

Jenny Fu
xf89@cornell.edu
Cornell University
Ithaca, NY, USA

Malte F. Jung
malte.jung@cornell.edu
Cornell University
Ithaca, NY, USA

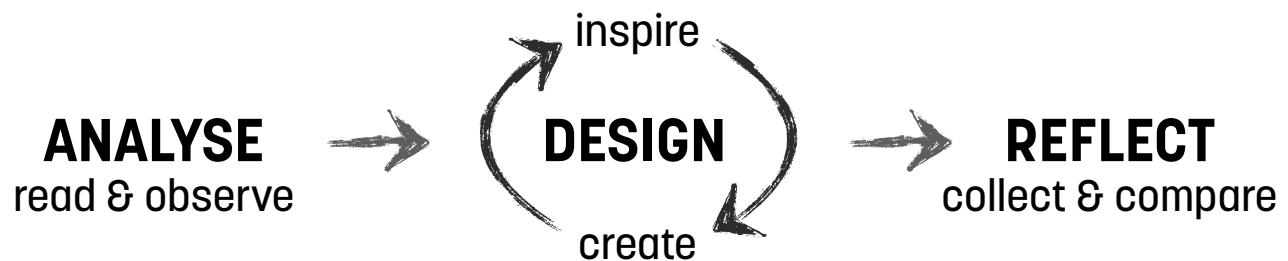


Figure 1: In traditional approaches, video analysis is applied in pre-studies before starting a new design (ANALYSE) or to evaluate an existing design (REFLECT). In this course we will focus on the actual prototyping stage (DESIGN), enabling participants to intertwine video analysis and interaction design in an iterative cycle.

ABSTRACT

In this course you will learn how to use video data for prototyping. The course provides hands-on training in working with video clips, including transcription and identification of relevant actions. You will familiarize with core interaction analytic concepts (grounded in ethnomethodology and conversation analysis) and will learn how to do an action-by-action analysis. Working on the design case of everyday interaction with automatic doors, you will learn how video interaction analysis can be embedded in an iterative design process.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design process and methods; HCI design and evaluation methods; Empirical studies in HCI.**

KEYWORDS

interaction analysis, interaction design, prototyping, video analysis

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1 BENEFITS

Designing for social and collaborative interaction can be challenging. Video recording interaction with technology helps to capture problems and creates a record that can be rewatched later. While there is a long history of video analysis and design (e.g. [5, 7, 13]) focus has typically been on design or just video analysis. This course aims to introduce course participants to a more integrated and iterative view of video interaction analysis and design.

The course teaches how video can be used to develop prototypes and how prototypes can be iteratively evaluated through video interaction analysis. The interaction analytic approach gives detailed insights into how human interaction is organized in detail and enables documentation of how people communicate in everyday encounters and in real time. Focus will be particularly on embodied actions, visual behavior and (non-lexical) sounds, as well as the timing of these, as they can become crucial design resources. The course will teach basic skills in using video data to identify such resources and how to prototype based on transcribed video data.

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2 INTENDED AUDIENCE & PREREQUISITES

The course is intended for a general audience that wants to know more about video interaction analysis, especially how moment-by-moment analysis can be embedded in a design process. Previous experience with video analysis is not required but may be an advantage. Participants should be able to download and install software for video playback and analysis. Prior experience with iterative prototyping may be beneficial.

3 COURSE CONTENT

The course introduces participants to the following topics:

- (1) Fundamentals of interaction analysis and its roots in ethnomethodology and conversation analysis, introducing concepts such as turn-taking [10, 11], sequences and projection [12], accountability [2], recipient design [10, 11] and multimodality [8].
- (2) The interaction analytic way of looking: identifying relevant sequences and actions in the video data
- (3) Software for frame-by-frame analysis and repeated watching
- (4) Visual/multimodal transcription
- (5) Using transcribed video for prototyping: storyboards and reenactments
- (6) Evaluation of prototypes through video interaction analysis

The course will be presented in four separate sessions. Each session will start with a short introduction to the theme by an instructor. Subsequently, students will split into small groups and work on hands-on exercises. Through the lectures and exercises, students will learn to

- identify key moments in video
- choose suitable software for video analysis
- formulate an analytical description of what is happening in a video snippet
- produce a basic transcript
- relate observations to fundamental analytic concepts in interaction analysis (rooted in the ethnomethodology & conversation analysis)
- develop prototypes based on short transcripts
- evaluate interaction with a prototype using video interaction analysis

4 PRACTICAL WORK: REDESIGNING AUTOMATIC DOORS

We will use a project to re-design automatic door interactions as a case to introduce and practice key concepts. The project is designed to provide a canvas for video based observation and interaction prototyping techniques.

4.1 Design Case

We chose automatic doors as project focus for several reasons. Since Don Norman's "Design of Everyday Things" [9], doors (e.g. "Norman Doors") have taken a special place in interaction design history as a case for discussions about interaction design. Automated doors are a particularly interesting technology to focus on because they have been introduced already during the 60s¹ and have been in use

without much change in their interaction design over the years. Finally, automatic doors are interesting as a simple one-degree-of-freedom robot to explore interaction design concepts for physical automated systems. In the words of Wendy Ju and Leila Takayama [6]:

"Automatic doors exemplify the possibilities and pitfalls of designing approachable interactive systems. We have all experienced the welcome, convenience, and ease of having doors sweep open before us as we draw near a building. Automatic doors are common enough to be conventional, invisible—almost unremarkable—and yet still they suffer from interaction problems. Any sustained observation of a building employing automatic doors will reveal numerous breakdowns: people have difficulty distinguishing automatic doors from non-automatic doors; people inadvertently trigger the doors without meaning to; people walk toward the door too quickly, or not quickly enough; people frustrated in their attempts to trigger the door before or after regular hours."

During the course we will move from observing interaction with automatic doors to analyzing and transcribing interaction with these doors with the purpose of redesigning them building on human practices for moving around these doors.

4.2 Practical Work

Course participants will be encouraged to observe interaction with automatic doors prior to the course and may record their own video snippets if they want to. At the beginning of the course, we will provide course participants with a video clip (about 30 seconds) which they will repeatedly watch and work on during the course. After students have completed a transcription exercise, we will make a full reference transcription available. The transcript will form the basis for the prototyping exercise, in which students will work on storyboards and enactments. Interaction with the prototypes will be video recorded and these video clips will be analysed during the final session of the course.

5 INSTRUCTOR BACKGROUND

- **Hannah Pelikan** is a PhD candidate in language and culture at Linköping University. In her research she combines video interaction analysis (ethnomethodology & conversation analysis) with research through design approaches. She has published on human-robot interaction at CHI, HRI and CSCW and co-organized workshops at HRI'21 and ICSR'20. She received her Masters in Interaction Technology from University of Twente, NL and teaches students from various backgrounds, including computer science.
- **Yoyo Tsung-Yu Hou** is a PhD student in Information Science at Cornell University. His research focuses on social interaction in Human-Robot Interaction and Human-Agent Interaction, especially on how social power theories can be used as a framework for designing interactive agents. Prior his PhD, he worked in industry as a UX designer for home robots with his dual backgrounds in psychology and product design.

¹https://en.wikipedia.org/wiki/Automatic_door



Figure 2: Automated doors open when one approaches them, opened by an invisible hand. However, in practice they often do not operate so smoothly, and people may experience all sorts of breakdowns.

- **Jenny Fu** is a PhD student in Information Science at Cornell University. Her research interests center on exploring how robots and AI agents can influence people's interaction between each other and applying social psychological theories to design conversational agents. Prior to Cornell, she received her undergraduate degree in Cognitive Science from Brown University.
- **Leelo Keevallik** is a professor in language and interaction at Linköping University. She has worked on the interface between grammar and embodiment, targeting the systematic use of language, sound, and movement in various activities involving co-present humans. Among other things, she has studied how vocalizations are used to coordinate manual work tasks and dance instruction. She has recently published a handbook on multimodal interaction analysis [1].
- **Mathias Broth** is professor in language and interaction at Linköping University. Taking particular interest in the embodied, emplaced and mediated aspects of situated human interaction, he has studied activities such as TV-production and driver training as these are interactionally accomplished in real time. He has recently published a handbook on multimodal interaction analysis [1].

- **Malte Jung** is an associate professor in information science at Cornell University. With a focus on moment-to-moment interactions, he has studied design teams as well teams that work with robots and other automated systems in order to design systems that help teams develop better interactions.

6 RESOURCES

For those who would like to deepen their knowledge in how to work with video, a range of handbooks in English is available. Some are walking through practical steps from picking the right recording equipment to producing first transcripts [3], while others focus on the analytic procedures [4]. Jordan and Henderson [5] give a general introduction to video analysis. Literature on specific interaction analytic topics as well as information on courses that introduce ethnomethodology and conversation analysis can be found at <http://emcawiki.net>.

7 ACCESSIBILITY

We will accommodate for the specific needs of the course participants based on consultation with them. The lecture slides will be available in advance so that participants can familiarize with the contents at their own pace before the course. We will take specific

needs into account when splitting participants into smaller groups with the goal to encourage an inclusive and enabling atmosphere in which we can cater to the needs of each individual student.

ACKNOWLEDGMENTS

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